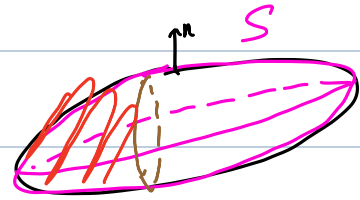
边值.

1. $\varphi|_s = u_1(s)$
2. $\frac{\partial\varphi}{\partial n}\Big|_s = u_2(s)$
3. $\frac{\partial\varphi}{\partial n}\Big|_{s_1} = u_3(s)$, $\varphi|_{s_2} = u_4(s)$



直观

S $\nabla^2\varphi=-\frac{\rho}{\varepsilon}$ 边界条件

φ 确定

$\varphi|_s$

$E=-\nabla\varphi$

$\nabla\cdot E=\frac{\rho}{\varepsilon} \Rightarrow \boxed{\nabla^2\varphi=-\frac{\rho}{\varepsilon}}$

φ的变化规律

静电场唯一性定理

问题
$$\begin{cases}\nabla^2\varphi=-\frac{\rho}{\varepsilon}\\ \varphi|_s=u(s),\quad \frac{\partial\varphi}{\partial n}\Big|_s=u(s)\end{cases}$$

$$\begin{cases}\nabla^2\varphi_1=-\frac{\rho}{\varepsilon},\quad \nabla^2\varphi_2=-\frac{\rho}{\varepsilon}\\ \varphi_1|_s=\varphi_2|_s,\quad \frac{\partial\varphi_1}{\partial n}\Big|_s=\frac{\partial\varphi_2}{\partial n}\Big|_s\end{cases} \Rightarrow \varphi_2=\varphi_1$$

令 $\Phi=\varphi_2-\varphi_1$　　$\nabla^2\Phi=\nabla^2\varphi_1-\nabla^2\varphi_2=0$

$$\begin{cases}\nabla^2\Phi=0\\ \Phi|_s=0\end{cases}$$

① $\Phi|_s=0$　一

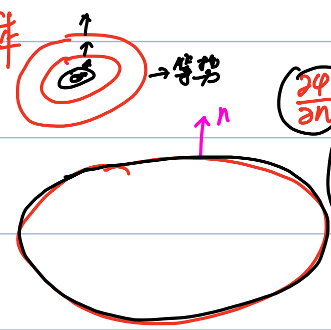
② $\nabla\Phi=0$　二

导体　等势

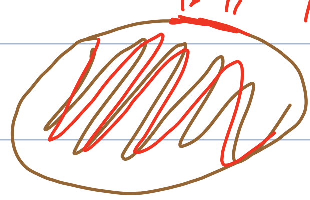
$\frac{\partial\varphi}{\partial n}$　梯度

等势体

$\nabla\varphi$



$\varphi_2=\varphi_1$　　$\varphi_2-\varphi_1=0$



$\oint_s \vec{f}\,ds=\iiint_V \nabla\cdot\vec{f}\,dV$

$\Phi\nabla\Phi$.

① $\Phi=0$,　$\nabla\cdot\nabla\Phi$, $\nabla^2\Phi=0$

② $\nabla\Phi=0$

$\oint \Phi\nabla\Phi\,ds=0$

$=\iiint \nabla\cdot(\Phi\nabla\Phi)\,dV=\iiint (\nabla\Phi)^2+\cancel{\Phi\nabla^2\Phi}=0$

$\varphi=\varphi_1$

静电场唯一性定理

$$= \iiint (\nabla \Phi)^2 dV = 0$$

$$\boxed{\nabla \Phi = 0}$$

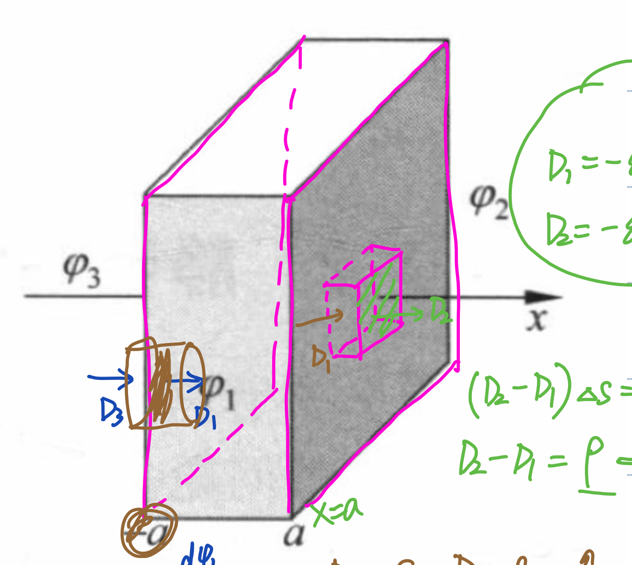
$$\left(\frac{\partial \Phi}{\partial x}, \frac{\partial \Phi}{\partial y}, \frac{\partial \Phi}{\partial z}\right) = 0.$$

$$\Downarrow$$

$\Phi =$ 常数 C

$$\boxed{\varphi_2 - \varphi_1 = 0}$$

例题 4-1 已知在直角坐标系下真空中电荷密度分布为 $\rho = \begin{cases} x, & |x| \leqslant a \\ 0, & |x| > a \end{cases}$，如图 4-1 所示。求空间中的电位分布。



$$D_1 = -\varepsilon_0 \frac{d\varphi_1}{dx}$$
$$D_2 = -\varepsilon_0 \frac{d\varphi_2}{dx}$$

$(D_2 - D_1)\Delta S = q$

$D_2 - D_1 = \rho = a$

$$\begin{cases} \nabla \cdot E = \dfrac{\rho}{\varepsilon_0} \\ E = -\nabla\varphi \end{cases}$$

$$\nabla^2 \varphi = -\frac{\rho}{\varepsilon_0}$$

$$\frac{d^2\varphi}{dx^2} = -\frac{\rho}{\varepsilon_0}$$

$$\rho = \begin{cases} x, & |x| \le a \\ 0, & |x| > a \end{cases}$$

$E = 0$

$$D_1 = -\varepsilon_0 \frac{d\varphi_1}{dx}$$
$$D_3 = -\varepsilon_0 \frac{d\varphi_3}{dx}$$

$D_1 \cdot \Delta S - D_3 \Delta S = q$

$D_1 - D_3 = \rho = -a$

(1) $|x| \le a$

$$\frac{d^2\varphi}{dx^2} = -\frac{x}{\varepsilon_0} \rightarrow \frac{d\varphi}{dx} = -\frac{x^2}{2\varepsilon_0} + C_1$$

(2) 当 $x > a$. $\dfrac{d^2\varphi}{dx^2} = 0 \rightarrow \dfrac{d\varphi}{dx} = C_1$

(3) 当 $x < a$ $\dfrac{d^2\varphi_3}{dx^2} = 0 \rightarrow \dfrac{d\varphi_3}{dx} = C_5$

$\varphi_3 = C_5 x + C_6$.

$(|x| \le a)$

$x=0,\ \varphi=0 \Rightarrow \boxed{C_2=0}$.

$\left.\frac{\partial\varphi}{\partial x}\right|_{x\to\pm\infty}=0 \Rightarrow C_3=C_5=0$.

$\varphi_1 = -\frac{x^3}{6\varepsilon_0}+C_1x$ $\quad |x|<a$

$\varphi_2 = +C_4$ $\quad x>a$

$\varphi_3 = +C_6$ $\quad x<-a$

$$\left\{\begin{array}{l}
\varphi_1=\varphi_2\Big|_{x=a} \\
\varphi_1=\varphi_3\Big|_{x=-a} \\
\left(-\varepsilon_0\cancel{\frac{d\varphi_3}{dx}}+\varepsilon_0\frac{d\varphi_1}{dx}\right)\Big|_{x=a}=a \\
\left(-\varepsilon_0\frac{d\varphi_1}{dx}+\varepsilon_0\frac{d\varphi_3}{dx}\right)\Big|_{x=-a}=-a
\end{array}\right.$$

$\rightarrow\ \varphi_1\big|_{x=a} = -\frac{a^3}{6\varepsilon_0}+C_1a = C_4$

$\rightarrow\ \varphi_1\big|_{x=a} = +\frac{a^3}{6\varepsilon_0}-C_1a = C_6$

$\rightarrow\ \varepsilon_0\left(-\frac{a^2}{2\varepsilon_0}+C_1\right)=a$

$\rightarrow\ C_1-\frac{a^2}{2\varepsilon_0}=\frac{a}{\varepsilon_0}$

$$\left\{\begin{array}{l}
C_1=\frac{a}{\varepsilon_0}+\frac{a^2}{2\varepsilon_0} \\
C_4=-\frac{a^3}{6\varepsilon_0}+\frac{a^2}{\varepsilon_0}+\frac{a^3}{2\varepsilon_0}= \\
C_6=
\end{array}\right.$$

$$\left\{\begin{array}{l}
\varphi_1=-\frac{x^3}{6\varepsilon_0}+\left(\frac{a}{\varepsilon_0}+\frac{a^2}{2\varepsilon_0}\right)x \\
\varphi_2=-\frac{a^3}{3\varepsilon_0}+\frac{a^2}{\varepsilon_0} \\
\varphi_3=\frac{a^3}{3\varepsilon_0}-\frac{a^2}{\varepsilon_0}
\end{array}\right.$$

例题 4-2 在无限大、电位为零的导体平板上方垂直放置一个无限长、张角为 2α 的导电圆锥，电位为 V，如图 4-2 所示。求导电圆锥与导电平板之间区域中的电位。

解 由题意可知，该种结构具有轴对称性。选用球坐标系，显然电位与方位角 ϕ 无关；又由于无限大的电位边界与径向矢量长度 r 无关，因此电位仅是极角 θ 的函数，故电位满足的拉普拉斯方程简化为

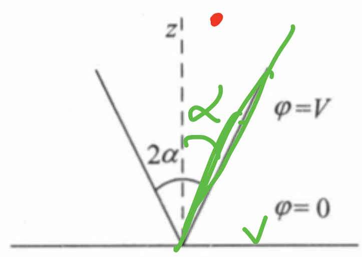


图 4-2 导电圆锥

对称 轴

① φ? → 无关.

$\nabla^2\varphi=0$

↓

② 方程？ $\nabla^2\varphi=0$

③ r 有关？ 无关

$$\cancel{\frac{1}{r^2\sin\theta}}\frac{d}{d\theta}\left(\sin\theta\frac{d\varphi}{d\theta}\right)=0$$

$$\frac{d}{d\theta}\left(\sin\theta\frac{d\varphi}{d\theta}\right)=0$$

$$\Downarrow$$

$$\sin\theta\frac{d\varphi}{d\theta}=C_1$$

$$\Downarrow$$

$$\frac{d\varphi}{d\theta}=\frac{C_1}{\sin\theta}$$

$$d\varphi=\frac{C_1}{\sin\theta}d\theta.$$

$$\varphi=\int\frac{C_1}{\sin\theta}d\theta.$$

$$=\int\frac{C_1\sin\theta d\theta}{\sin^2\theta}$$

C

$$=\int\frac{C_1(-d\cos\theta)}{1-\cos^2\theta}$$

$$=\int\frac{C_1 d\cos\theta}{(1+\cos\theta)(1-\cos\theta)}$$

$$=C_1\int\frac{1}{2}\left(\frac{1}{1+\cos\theta}+\frac{1}{1-\cos\theta}\right)d\cos\theta$$

$$=C_1\ln\frac{1+\cos\theta}{1-\cos\theta}+C_2$$

$$\cos\theta=2\cos^2\frac{\theta}{2}-1=1-2\sin^2\frac{\theta}{2}$$

+

$$= C_1 \ln \frac{2\cos^2\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} + C_2$$

$$= C_1 \ln \frac{1}{\tan^2\frac{\theta}{2}} + C_2$$

$$= C_1(-2)\ln \tan \frac{\theta}{2} + C_2$$

$$= C_1 \ln\left(\tan\frac{\theta}{2}\right) + C_2$$

$$\varphi(\theta) = C_1 \ln\left(\tan\frac{\theta}{2}\right) + \cancel{C_2}$$

① $\theta = \frac{\pi}{2}$ $\quad C_1 \ln\left(\tan\frac{\pi}{4}\right) + C_2 = 0 \Longrightarrow C_2 = 0$

$\tan\frac{\pi}{4} = 1$, $\quad \ln(1) = 0$

② $\theta = \alpha$. $\quad C_1 \ln\left(\tan\frac{\alpha}{2}\right) = V \qquad C_1 = \frac{V}{\ln\left(\tan\frac{\alpha}{2}\right)}$

$$\varphi(\theta) = \frac{V}{\ln\left(\tan\frac{\alpha}{2}\right)} \cdot \ln\left(\tan\frac{\theta}{2}\right) \quad \checkmark$$